MST: A Multi-Scale Transformer Framework With Cross-Scale Token Fusion for Automatic Modulation Recognition

Jingbo Zhang, *Member, IEEE*, Shaoqian An, Fanxiang Meng, and Qingshuo Liu

***Abstract*—Automatic modulation recognition which is a core enabling technology for cognitive radio systems, faces considerable challenges in robust feature extraction under non-ideal environments. Under low signal-to-noise ratio (SNR) conditions and modulation parameter mismatches, conventional deep learning-based methods exhibit markedly degraded discriminative feature representation capability. This letter presents a multi-scale Transformer (MST) framework that significantly enhances feature extraction from raw signal sequences through parallel multi-resolution analysis. We decompose signals into three complementary scales using varying convolutional kernel sizes, enabling comprehensive analysis of global features, local characteristics, and transient details across amplitude, phase, and frequency domains. Subsequently, an effective cross-scale token fusion mechanism is developed to integrate information across different scales. Furthermore, we employ knowledge distillation to compress the model, nearly halving its size while maintaining performance. Experimental results show that under challenging conditions combining low SNR, large carrier frequency offsets, and significant sampling rate offsets, the proposed MST framework demonstrates superior performance compared to other state-of-the-art methods.**

***Index Terms*—Automatic modulation recognition, transformer, carrier frequency offsets, sample rate offsets, deep learning.**

I. Introduction

AUTOMATIC modulation recognition (AMR) is a very interesting research direction in cognitive radio, aiming to accurately identify the modulation type of wireless signals in non-cooperative communication systems [1]. Traditional AMR methods can be categorized into likelihood-based approaches and feature-based approaches. The former approaches formulate AMR as a composite hypothesis testing problem, in which modulation pattern recognition is performed through the construction and evaluation of likelihood ratio test functions [2]. However, these approaches require specific prior information, which is often difficult to obtain in non-cooperative communication environments. In comparison, feature-based methods rely on domain-specific expertise to construct discriminative signal features for classification decisions [3]. The main challenge is how to extract highly separable feature vectors from time series signals, which need to maintain robust classification performance in high-dimensional spaces.

With the advancements in computer technology and the establishment of unified datasets [4], deep learning has emerged as the dominant approach for AMR tasks due to its powerful feature extraction capacity [5]. Early studies predominantly employed Convolutional Neural Networks (CNNs) as the core architecture [6], focusing on extracting spatial features from signals. To better capture temporal dependencies, researchers subsequently proposed hybrid models combining CNNs with Recurrent Neural Networks (RNNs) [7], [8], [9]. However, the ability of RNNs to model long sequences is inherently constrained by factors such as the sequential computation mechanism and the vanishing gradient problem. This critical bottleneck was overcome with the introduction of the Transformer architecture [10], whose self-attention mechanism can directly model global dependencies. This breakthrough has led to the development of high-performance models such as FEA-T [11].

However, these approaches still face new challenges in practical applications. For instance, in non-cooperative environments where the receiver lacks prior information about the transmitter, large carrier frequency offsets (CFOs) and significant sampling rate offsets (SROs) may occur during received signal preprocessing [12]. This condition is referred to as modulation parameter mismatch (MPM). Through analysis, we found that large CFOs cause radial distortion in the modulation domain, while significant SROs induce scaling distortion in the time domain. These distortions, when combined with the influence of noise, further degrade the detection performance of conventional single-scale models.

To this end, inspired by CrossViT [13], we propose a multi-scale Transformer (MST) framework. The main contributions are as follows:

- We employ multi-scale convolutional kernels to process signal amplitude, phase, and frequency features in parallel, while a group convolution residual module captures feature variations synchronously. During the sliding process, convolutional kernels of different scales can focus on the feature correlations across various dimensions of the signal symbols. This architecture allows the MST network to achieve richer feature representations.
- We construct a Triple Transformer network and propose a cross-scale token fusion mechanism (CS-TFM) to mitigate the influence of MPM. This enhancement improves the cross-attention module's signal feature representation capability during multi-scale feature fusion, thereby strengthening the network's overall modeling capacity.
- We propose a knowledge distillation (KD) method designed specifically for the CS-TFM module. This

Received 19 August 2025; accepted 20 September 2025. Date of publication 24 September 2025; date of current version 12 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62071080 and Grant 61501078. The associate editor coordinating the review of this article and approving it for publication was C. H. Liu. *(Corresponding author: Jingbo Zhang.)*

The authors are with the Information Science and Technology College and the National Engineering Research Center of Maritime Navigation System, Dalian Maritime University, Dalian 116026, Liaoning, China (e-mail: zhang_jingbo@dlmu.edu.cn; asq@dlmu.edu.cn; mfx3128953318@dlmu.edu.cn; liuqingshuo_dlmu@dlmu.edu.cn).

Digital Object Identifier 10.1109/LWC.2025.3614235

2162-2345 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on November 12,2026 at 07:53:49 UTC from IEEE Xplore. Restrictions apply.

method can effectively transfer practical knowledge from the teacher model to the student model [14].

Furthermore, we conduct validation tests on the Da-RxMism dataset with MPM interference [12]. The results demonstrate that our model outperforms other state-of-the-art approaches.

II. Signal Model

We mainly considered the signal reception process in a Single-Input Single-Output (SISO) system. The received continuous signal $r(t)$ can be expressed mathematically as:

$$r(t) = [s(t) * h(t)]e^{j(2\pi\Delta f t+\phi)} + n(t), \tag{1}$$

where $s(t)$ denotes the transmitted symbol, $h(t)$ represents the wireless channel impulse response, $n(t)$ is the additive white Gaussian noise, Δf is the CFO primarily caused by LO frequency mismatches between tx/rx, and ϕ is the phase shift.

Subsequently, $r(t)$ is processed by the receiver and undergoes sampling, which can be represented as:

$$r_s(t) = r(t)\delta_T(t) = \sum_{n=-\infty}^{+\infty} r(nT)\delta(t-nT), \tag{2}$$

where $\delta_T(t)$ is a periodic impulse train with period T.

In general, thermal jitter or clock timing anomalies in the receiver can introduce offsets during the sampling process. Such offsets are as follows:

$$T = \frac{1}{f_s + \Delta f_s}, \tag{3}$$

where f_s is the sampling rate and Δf_s is SRO.

In this letter, we primarily investigate the MPM scenario, where the signal is plagued by high-intensity noise, large CFOs (with $\Delta f/f$ up to ± 0.5), and significant SROs (with $\Delta f_s/f_s$ up to ± 0.5). Additionally, we store $r_s(t)$ as I/Q sequences and then process them into APF format.

III. Proposed Method

A. Proposed Multi-Scale Transformer

The whole architecture of the proposed MST is drawn in Fig. 1. The MST incorporates a comprehensive three-stream architecture, employing tokens that capture long-term signal trends (LTST), local inter-symbolic relations (LISR), and transient spectral details (TSD) for the purpose of feature extraction. More specifically, we utilize three convolutional layers, each tailored to emphasize one of these aspects and characterized by unique kernel sizes. The kernel sizes are set to 4χ, 2χ, and χ, where χ represents the number of sampling points per symbol and serves as a hyperparameter initialized based on prior information. Meanwhile, the stride is configured as half the length of the corresponding convolutional kernel. This setting facilitates obtaining a larger feature receptive field. Afterwards, the embedding layer and the positional encoding layer process the three feature streams, which can be represented mathematically as:

$$\mathbf{X}_\lambda = \left[\mathbf{x}^0_{Cls}; \mathbf{x}^1_\lambda \mathbf{W}_\lambda; \mathbf{x}^2_\lambda \mathbf{W}_\lambda; \cdots; \mathbf{x}^N_\lambda \mathbf{W}_\lambda\right] + \mathbf{W}_{pos}, \tag{4}$$

where $\lambda \in \{ltst, lisr, tsd\}$ denotes three scales, $\mathbf{X}_\lambda$ represents tokens at these three scales.

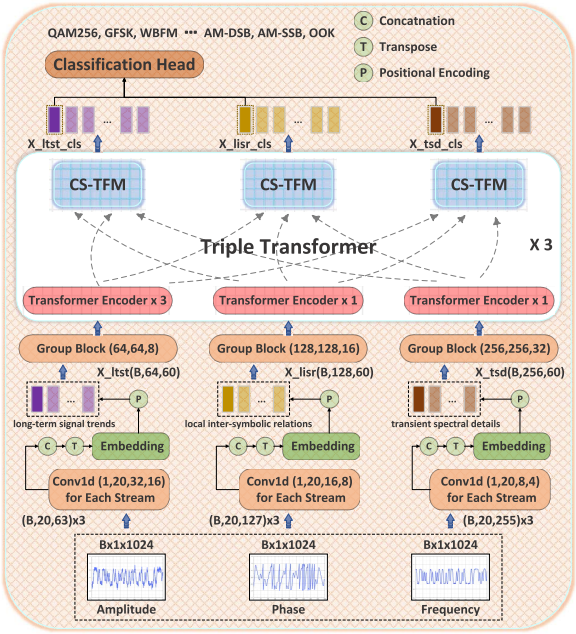


Fig. 1. The structure of the multi-scale Transformer.

In the feature extraction and fusion phase, we employ a Triple-Transformer module. The $\mathbf{X_{ltst}} \in \mathbb{R}^{B\times 64\times 60}$ serves as the primary branch for feature extraction, passing through three Transformer encoder layers to capture global features. Meanwhile, $\mathbf{X_{lisr}} \in \mathbb{R}^{B\times 128\times 60}$ and $\mathbf{X_{tsd}} \in \mathbb{R}^{B\times 256\times 60}$ act as complementary branches, each undergoing only one for fine-grained feature extraction. This design aims to enhance the module's global modeling capability while effectively balancing the model's parameter scale. And the feature fusion is achieved through the CS-TFM. Note that in our Transformer encoder layers, the Feed-Forward Networks are enhanced by incorporating a plug-and-play module, specifically the Swish-Gated Linear Unit [15]. It can be expressed as:

$$SwiGLU\left(x^{att}\right) = Swish\left(x^{att}\mathbf{W} + \mathbf{b}\right) \otimes \left(x^{att}\mathbf{V} + \mathbf{c}\right), \tag{5}$$

where x^{att} is the attention-weighted feature, $\otimes$ is the element-wise product. In addition, the $Swish(\cdot)$ function satisfies:

$$Swish(x) = \frac{x}{1+e^{-x}}. \tag{6}$$

In addition, to further optimize the feature representation, we add two GroupBlocks before the features enter the Triple-Transformer module for pre-extraction of local features. The architecture can be described as follows:

$$Grp(x^e) = \sigma(\phi_1(1) \circ \sigma(\phi_3(g) \circ \sigma(\phi_1(1) \circ x^e)) + x^e), \tag{7}$$

$$\mathbf{Y}_\lambda = Grp \circ Grp \circ \mathbf{X}_\lambda, \tag{8}$$

where x^e represents the embedded feature, $\phi_j(g)$ denotes the 1D convolution with a kernel size of j and g groups, $\sigma(\cdot)$ is the ReLU activation function.

In the final classification stage, after comparing different classification heads, we ultimately chose to use the arithmetic mean of the classification results from three branches.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on November 12,2026 at 07:53:49 UTC from IEEE Xplore. Restrictions apply.

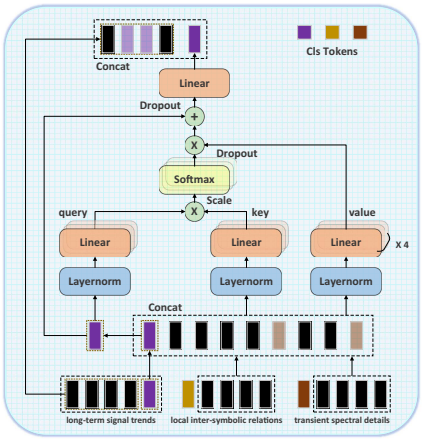


Fig. 2. The token fusion mechanism for LTST branch.

B. Cross-Scale Token Fusion Mechanism

Prior to the integration of the proposed CS-TFM, the three branches of MST independently extract features from the raw signal, capturing distinct scale-specific representations. To facilitate collaborative learning across branches while preserving their individual feature dominance, this letter proposes a token fusion strategy tailored for multi-feature stream optimization. The strategy leverages the semantic differentiation between the Cls token and Patch tokens: specifically, the Cls token encodes branch-level prototypical features, whereas Patch tokens capture derived features from local regions.

As illustrated in Fig. 2, the mechanism achieves feature fusion through cross-branch token interactions. Taking the LTST branch as an example, the fusion process comprises the following steps:

1) The LTST branch aggregates patch tokens from the LISR and TSD branches to acquire derived features, and then concatenates its own Cls token with the collected cross-branch patch tokens to form a merged representation, as shown in Eq. (9) below.

$$\mathbf{X}'^{ltst} = \left[\mathbf{x}_{Cls}^{ltst}; \mathbf{x}_{patch}^{lisr}; \mathbf{x}_{patch}^{tsd}\right]. \tag{9}$$

2) An attention module is employed to facilitate information exchange across branches. The query, key, and value weights are derived as illustrated in the figure. Following the attention computation, the final module yields an enhanced Cls token that not only incorporates complementary information from other branches but also preserves the branch-specific semantics. Mathematically, this process can be formulated as:

$$\begin{cases} \mathbf{Q} = LN\left(\mathbf{x}_{Cls}^{ltst}\right)\mathbf{W}_q, \\ \mathbf{K} = LN\left(\mathbf{X}'^{ltst}\right)\mathbf{W}_k, \\ \mathbf{V} = LN\left(\mathbf{X}'^{ltst}\right)\mathbf{W}_v, \\ \mathbf{A}_j = softmax\left(\frac{\mathbf{Q}_j\mathbf{K}_j^T}{\sqrt{d_k}}\right)\mathbf{V}_j, \\ \mathbf{Z} = [\mathbf{A}_1; \cdots; \mathbf{A}_4] + \mathbf{x}_{Cls}^{ltst}, \end{cases} \tag{10}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, $\mathbf{W}_v$ are learnable parameters, and $\mathbf{A}_j$ represents the attention head in the attention mechanism.

3) Map the enhanced Cls token back to its original branch, and subsequently, permit the patch tokens to update local derived features based on the new Cls token, which can be expressed as follows:

$$\mathbf{Output} = \left[\mathbf{ZW} + \mathbf{b}; \mathbf{x}_{patch}^{ltst}\right]. \tag{11}$$

This design effectively captures the inter-branch feature dependencies, promoting the complementary utilization of multi-scale features while preserving the distinctiveness of each branch's representations. Meanwhile, a comparable process is performed in parallel in other branches, as shown in Fig. 1.

C. Knowledge Distillation

To reduce the parameter overhead of the three-branch network, we employ a simplified version of the MST-S model and utilize knowledge distillation technology, using the MST model as the teacher to guide the MST-S. Specifically, MST-S modifies the network structure illustrated in Fig. 1 by reducing the number of Conv1d channels from 20 to 16 and decreasing the number of triple-transformer modules from 3 to 2. Due to the computational characteristics of the Transformer architecture, such structural adjustments significantly reduce the overall parameter count of the model.

Considering the incompatibility between the traditional approach of adding a distillation token in Transformer models and the CS-TFM module, which relies on cross-attention, this research proposes a specially designed knowledge distillation loss function. Specifically, the Cls tokens output by the last two CS-TFM modules of the teacher model are selected as soft labels. These soft labels are then used to compute the KL divergence with the Cls tokens output from the corresponding two CS-TFM modules of the student model, thereby constructing a prototypical features loss term. Subsequently, this loss is combined with the hard loss and soft loss output by the fc layer to form the final hybrid loss function, as follows:

$$Potyloss = \sum_{n \in C_{list}} KL(Cls_t, Cls_s, T), \tag{12}$$

$$Hardloss = L_{ce}(y_l, y_s), Softloss = L_{ce}\left(\frac{y_t}{T}, \frac{y_s}{T}\right), \tag{13}$$

$$Loss = a * Hardloss + b * Softloss + c * Potyloss, \tag{14}$$

where L_{ce} denotes the cross-entropy loss, T represents the temperature coefficient, y_l is the true label, y_s and y_t are the fc-layer outputs of the student and teacher networks respectively.

IV. Experimental Evaluation

A. Dataset and Experiment Settings

To comprehensively evaluate the model's performance under MPM scenarios, we adopted the open-source dataset (Da-RxMism) provided in [12]. During the experiments, data samples encompassing various modulation types at different Signal-to-Noise Ratios (SNRs) were divided into training, testing, and validation sets in a ratio of 5:2.5:2.5. In addition, all experiments were conducted remotely on servers equipped with NVIDIA CUDA-driven GPUs, such as 3090 models.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on November 12,2026 at 07:53:49 UTC from IEEE Xplore. Restrictions apply.

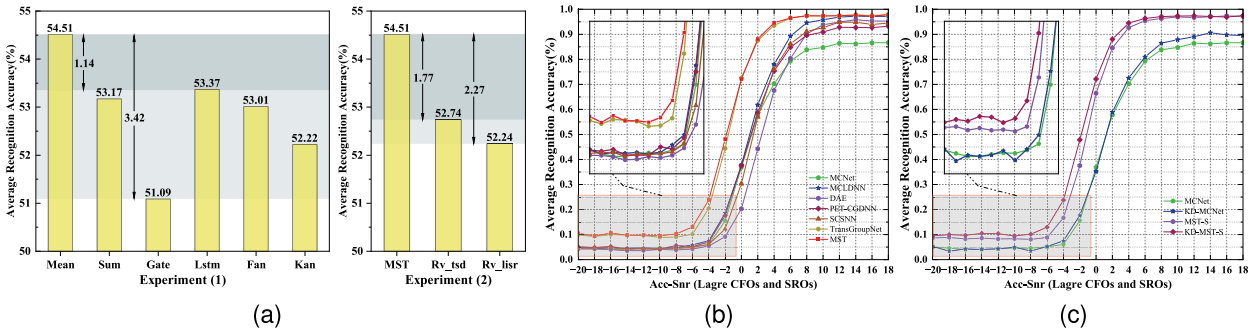


Fig. 3. (a) Control Experiment. (b) Comparison Experiment. (c) Distillation Experiment.

During the model training process, we adopted a two-phase training strategy. In the first phase, we selected the AdamW optimizer with an initial weight decay of 5×10^{-2}, and set the initial learning rate to 5×10^{-3}. If the validation loss failed to decrease within 5 epochs, the learning rate would be decayed by a factor of 0.8. Throughout this phase, the batch size was set to 512, and a total of 195 training epochs were conducted.

In the second phase, we initialized a new MST model using the best weights obtained from the first phase. Subsequently, we conducted an additional five training epochs on this model, setting the initial learning rate to 1×10^{-4}. Furthermore, for all experiments, we utilized the cross-entropy loss function with label smoothing of 0.1.

B. Ablation Study

To explore the optimal model architecture, we conducted four sets of comparative experiments.

1) Different Strategies to Use the Temporal Outputs of the Transformer Encoder Layer: In Transformer-based models, single-stream designs [11], [12], [14] often classify using the Cls token via an MLP. However, in our model, which adopts a three-stream design, effectively leveraging the outputs of the Transformer poses an intriguing question. In this control experiment, we examined 6 strategies for using the Cls token: Mean, Sum, and others (details below).

- Gate: $\phi(x) = sum(g_1'\cdot f(\mathbf{x}_{Cls}^{ltst}), g_2'\cdot f(\mathbf{x}_{Cls}^{lisr}), g_3'\cdot f(\mathbf{x}_{Cls}^{tsd}))$.
- Lstm: $\phi(x) = f(lstm(\mathbf{x}_{Cls}^{ltst}, \mathbf{x}_{Cls}^{lisr}, \mathbf{x}_{Cls}^{tsd}))$.
- Fan: $\phi(x) = f(fan(\mathbf{x}_{Cls}^{ltst}), fan(\mathbf{x}_{Cls}^{lisr}), fan(\mathbf{x}_{Cls}^{tsd}))$.
- Kan: $\phi(x) = mean(kan(\mathbf{x}_{Cls}^{ltst}), kan(\mathbf{x}_{Cls}^{lisr}), kan(\mathbf{x}_{Cls}^{tsd}))$.

Here, $f(\cdot)$ denotes the dense layer, while $fan(\cdot)$ and $kan(\cdot)$ are defined in [16] and [17], respectively. Additionally, g_j' is defined as $\frac{g_j}{g_1+g_2+g_3}$, where g_j is a learnable parameter.

In our study, as illustrated in Fig. 3(a), employing the Mean approach to process the temporal outputs of the Transformer encoder layer yielded the best performance. This strategy outperformed the second-best approach, which utilized Lstm, by a margin of 1.14% in accuracy. Moreover, it surpassed the worst-performing method, employing a Gate mechanism, by a significant 3.42% in accuracy. These results underscore the critical importance of treating the three branches equally during training. Upon examining the final weights of g_1', g_2', and g_3' for the Gate strategy (specifically, 0.4068, 0.1922, and 0.4010 respectively), we found that its inferior performance was attributed to an undue neglect of LISR branch.

2) CS-TFM vs. Cross-Attention: To verify the effectiveness of CS-TFM, we pruned the TSD and LISR feature streams in MST to construct two simplified two-branch networks, in which cross-attention was employed to fuse their features. As depicted in Fig. 3(a), the Rv-tsd led to a decrease in accuracy by 1.77%, whereas the Rv-lisr resulted in a 2.27% accuracy drop. This fully demonstrates the superiority of CS-TFM, as well as the advantages of multi-scale feature fusion.

3) MST vs. MST without GroupBlock: In numerous Transformer-based models, the superiority of CNN with Transformer has been validated. And, we further explored this point and observed a 2.38% decline in the classification performance of MST when the GroupBlock module was removed.

Notably, we adopted a unique design for the number of groups within the GroupBlock. This design originates from the configuration of convolutional layers employed in the signal decomposition stage, as discussed in previous sections. Specifically, the grouping factor in the GroupBlock bears an inverse relationship with the kernel sizes used in the signal decomposition stage ($[32, 16, 8] \Rightarrow [8, 16, 32]$). Primarily, this design aims to enhance computing efficiency. Furthermore, the results of the experiments demonstrate that adopting other grouping methods leads to a 1-3% decline in performance.

4) Scale Control Experiment: In this particular study, we trained the MST, Rv_tsd, and Rv_lisr models using scale benchmarks $\chi = [4, 8, 12]$, respectively. Furthermore, the obtained accuracies were $Acc_{mst} = [52.76\%, 54.51\%, 53.01\%]$, $Acc_{rv_tsd} = [51.98\%, 52.74\%, 52.38\%]$, and $Acc_{rv_lisr} = [52.19\%, 52.24\%, 52.09\%]$. These results effectively validate the superiority of our proposed models. Among these models, the Rv_lisr model demonstrated the least variation in accuracy, indicating that different scale sizes have a relatively significant impact on the lisr branch. The lisr branch can more effectively capture inter-symbol feature relationships only when the scale benchmark χ is set to the number of sampling points in a symbol. Additionally, we attempted to construct a 4-branch model by introducing an additional scale benchmark $\chi = [24, 48]$. However, due to the excessive model size, overfitting occurred, ultimately leading to a decline in accuracy. We also explored the application of these two scales to our ltst branch, but the results were similarly unsatisfactory.

C. Comparison to State-of-the-Art

To explore the superiority of the proposed MST method, we conducted comparative experiments with several algorithms employed in this field over the past five years, including

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on November 12,2026 at 07:53:49 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EXPERIMENTS ON THE DA-RXMISM

Method	Flops(M)	Params(M)	Average Accuracy(%)
MCNet	**11.241**	0.127	40.76
MCLDNN	343.16	0.370	45.43
DAE	13.961	**0.015**	41.10
PET-CGDNN	71.924	0.076	43.50
SCSNN	451.24	9.545	43.20
TransGroupNet	41.518	0.585	53.80
MST	109.59	1.061	**54.51**

MCNet [6], MCLDNN [7], DAE [8], PET-CGDNN [9], SCSNN [18] and TransGroupNet [12]. As shown in Tab. I, on the Da-RxMism dataset, the proposed MST method achieved the best performance, with an accuracy improvement of 0.71% over the second-best method (TransGroupNet) and 13.75% over the worst-performing baseline (MCNet).

Furthermore, Fig. 3(b) elaborates on the performance of various algorithms under different SNR conditions on the Da-RxMism. The experimental results demonstrate that within the low SNR range of –20dB to –2dB, our proposed MST algorithm exhibited a significant advantage, clearly illustrating its excellent robustness against noise. Additionally, within the SNR range of 10dB to 18dB, the MST algorithm continued to outperform most algorithms, further validating its good robustness in terms of CFO and SRO. Notably, under high SNR conditions, the MCNet algorithm performed the worst, reflecting the significant negative impact of CFO and SRO on signal spatial characteristics, which is consistent with the consensus in the communications field. Since residual carriers introduce numerous harmonic interferences into the signal waveform, and large SROs result in significant waveform distortion, algorithms that solely rely on spatial features for classification (such as MCNet) are particularly affected.

D. Knowledge Distillation Experiment

We utilized the MST model as the teacher model, adopted the loss function designed in Section III-C, and trained the MST-S model following the previous strategy. Ultimately, it achieved an accuracy of 54.43% ($T = 5$,$a = 1$, $b = 0.7$, $c = 0.3$) Its parameter size was nearly halved (55.580M Flops, 0.668M Params). It confirms the effectiveness of the proposed knowledge distillation strategy, which enables robust transfer of MST knowledge and exhibits strong adaptability to cross-attention-like modules.

To further verify the effectiveness of cross-network MST knowledge transfer, we trained the lightweight MCNet as the student model. As illustrated in Fig. 3(c), the proposed knowledge distillation approach enhanced recognition accuracy by 1.13%. This finding demonstrates that complex networks can be effectively utilized during offline training to learn high-dimensional signal features. For industrial deployment, knowledge distillation provides a means of transferring this knowledge to lightweight networks, thus achieving improved recognition accuracy under practical constraints.

V. CONCLUSION

This letter presents a novel multi-scale Transformer framework with cross-scale token fusion, which is designed to enhance discriminative feature extraction from raw signal sequences and mitigate the impacts of MPM and noise interference. Experimental results show that the proposed framework outperforms the state-of-the-art approaches on the Da-RxMism datasets. We also design related experiments to prove the robustness of the framework to interference factors such as low SNRs, large CFOs, and significant SROs. Finally, we use a specific KD method to compress the model. Even though its size is nearly halved, it retains recognition accuracy. Moreover, we leverage it as a teacher model to guide the training of lightweight models, validating its industrial value.

ACKNOWLEDGMENT

This letter was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

REFERENCES

[1] S. Chang et al., "A fast multi-loss learning deep neural network for automatic modulation classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 9, no. 6, pp. 1503–1518, Dec. 2023.
[2] J. Zhang et al., "Cooperative modulation classification for multipath fading channels via expectation-maximization," *IEEE Trans. Wireless Commun.*, vol. 16, no. 10, pp. 6698–6711, Oct. 2017.
[3] N. Jafar et al., "Automatic modulation classification using modulation fingerprint extraction," *J. Syst. Eng. Electron.*, vol. 32, no. 4, pp. 799–810, Aug. 2021.
[4] T. J. O'Shea et al., "Over-the-air deep learning based radio signal classification," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 1, pp. 168–179, Feb. 2018.
[5] S. Chang, S. Huang, R. Zhang, Z. Feng, and L. Liu, "Multitask-learning-based deep neural network for automatic modulation classification," *IEEE Internet Things J.*, vol. 9, no. 3, pp. 2192–2206, Feb. 2022.
[6] T. Huynh-The et al., "MCNet: An efficient CNN architecture for robust automatic modulation classification," *IEEE Wireless Commun. Lett.*, vol. 24, no. 4, pp. 811–815, Apr. 2020.
[7] J. Xu et al., "A spatiotemporal multi-channel learning framework for automatic modulation recognition," *IEEE Wireless Commun. Lett.*, vol. 9, no. 10, pp. 1629–1632, Oct. 2020.
[8] Z. Ke and H. Vikalo, "Real-time radio modulation classification with an LSTM auto-encoder," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2021, pp. 4935–4939.
[9] F. Zhang et al., "An efficient deep learning model for automatic modulation recognition based on parameter estimation and transformation," *IEEE Commun. Lett.*, vol. 25, no. 10, pp. 3287–3290, Oct. 2021.
[10] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[11] Y. Chen et al., "Abandon locality: Frame-wise embedding aided transformer for automatic modulation recognition," *IEEE Commun. Lett.*, vol. 27, no. 1, pp. 327–331, Jan. 2023.
[12] R. Zeng et al., "Convolutional neural network assisted transformer for automatic modulation recognition under large CFOs and SROs," *IEEE Signal Process. Lett.*, vol. 31, pp. 741–745, 2024.
[13] C.-F. R. Chen et al., "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 347–356.
[14] D. Hou et al., "CLST: A convolutional transformer framework for automatic modulation recognition by knowledge distillation," *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 8013–8028, Jul. 2024.
[15] N. Shazeer, "Glu variants improve transformer," 2020, *arXiv:2002.05202*.
[16] Y. Dong et al., "Fan: Fourier analysis networks," 2024, *arXiv:2410.02675*.
[17] A. D. Bodner et al., "Convolutional Kolmogorov-Arnold networks," 2024, *arXiv:2406.13155*.
[18] T. Wei et al., "Adaptive multi-dimensional shrinkage block for automatic modulation recognition," *IEEE Wireless Commun. Lett.*, vol. 27, no. 11, pp. 2968–2972, Nov. 2023.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on November 12,2026 at 07:53:49 UTC from IEEE Xplore. Restrictions apply.